Rethinking Customer Experience: Predictive, Non-Intrusive, Always-On

RetailJarvis: From Data to Dialogue

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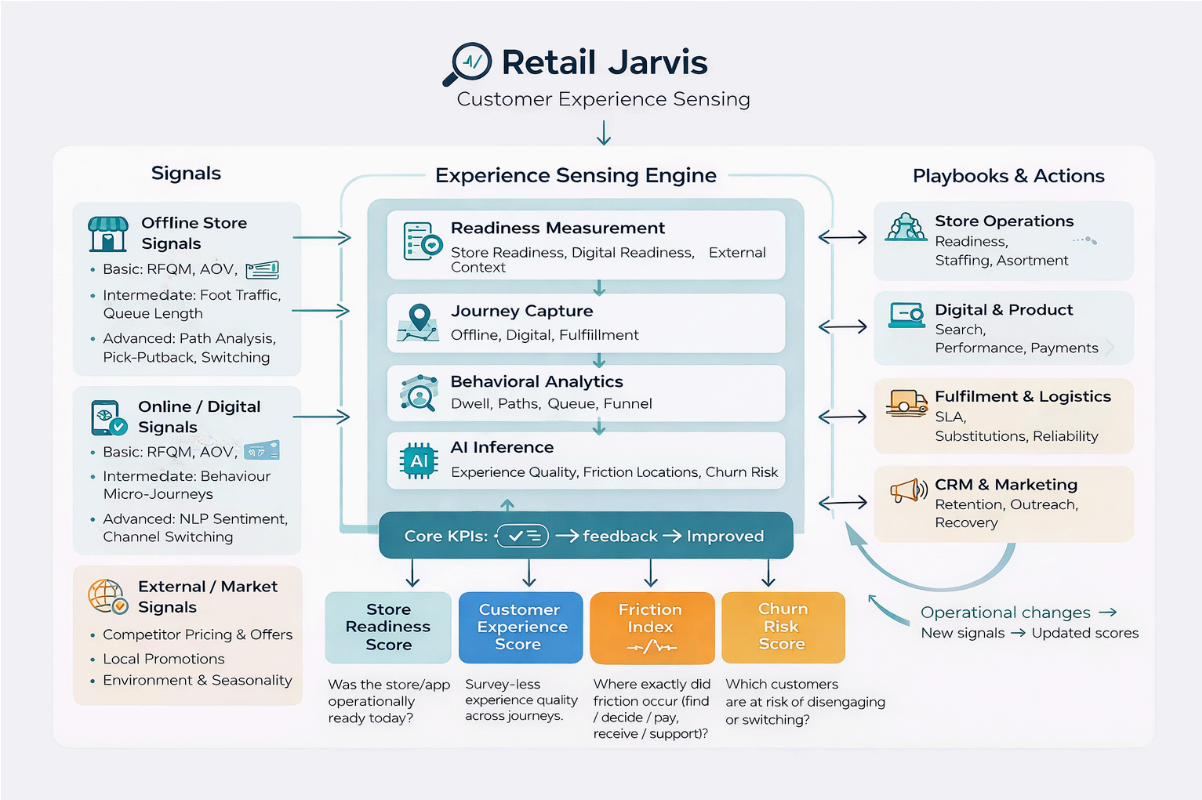

Conversational RetailVista

- In the past week, as we engaged with teams across businesses, it became evident that Reliance's expanding data ecosystem and the rapid growth of use cases demand a more scalable and intelligent way to interact with information.

SECTION 1 — FIRST PRINCIPLES: How We Understand Customer Experience Without Asking Them

- The foundation of this approach is simple: customer experience cannot be understood by asking customers. It must be understood by **observing what actually happens**, starting from the ground up — in our stores, in our digital journeys, and in the context customers operate in.

1. Start With Operational Reality, Not Customer Feedback

Before interpreting any behaviour or predicting any outcome, we must ensure the fundamentals of the shopping environment are working as intended. These basics directly shape how a customer feels:

- Temperature, lighting, cleanliness, and overall comfort
- Availability of products in expected locations
- Ease of navigating the store
- Staff responsiveness and readiness
- App stability, search performance, and checkout ease
- Delivery completeness and timeliness
- Competitor pricing and offers influencing expectations before a visit

If these conditions are not right, every downstream signal from customers becomes distorted.
So the model begins by measuring and fixing **basic readiness** — offline and online.

2. Experience Is Reflected in Behaviour Long Before It Is Ever Verbalised

Customers communicate satisfaction or dissatisfaction through their actions:

- Whether they continue browsing or exit quickly
- Whether they put items back
- Whether they abandon a queue
- Whether they return more frequently or slowly reduce visits
- Whether they retry payments or drop off
- Whether they complain publicly or silently switch channels
- Whether they react to competitor pricing by shifting baskets

These behavioural signals are far richer and more honest than any form-based feedback.
By reading these actions consistently, we gain a real understanding of experience without interrupting customers.

3. Build a View of Customer Experience Bottoms-Up, Not Top-Down

- A predictive experience system must be built in the right order:

A. Measure readiness

- Environmental, operational, digital, and competitive conditions that set the stage for experience.

B. Capture the journey

- What actually happened: movement, dwell, search quality, friction points, delivery outcomes.

C. Observe customer response

- Changes in purchase behaviour, returns, switching between online and offline, reaction to competitor pricing, and public sentiment.

D. Only then infer and predict

- Experience quality, friction hotspots, and future risk.
- This ensures the model is grounded, accurate, and reflective of what truly happens in the shopping environment — not assumptions, not surveys, and not sentiment sampling.

4. External Context Must Be Part of the Experience Model

- Customer expectations are shaped before they enter our stores or open the app.
  Signals such as:
  - Competitor pricing
  - Promotions in the local market
  - Social media sentiment
- Seasonal or environmental factors (heat, rainfall, festivals)…must be integrated so that the system interprets customer behaviour in its true context.

Summary of First Principles

1. **Fix and measure the basics first — readiness defines experience.**
2. **Observe actions, not opinions — behaviour reveals truth.**
3. **Build bottom-up — readiness → journey → behaviour → inference.**
4. **Incorporate external signals like competitor pricing and public sentiment.**

This establishes a solid foundation for everything that comes next.

How These Principles Lead to the Core Outputs

By applying these first principles consistently, the system produces four clear, actionable outputs:

1. Store Readiness Score (Offline)

- The foundation for interpreting all in-store behaviour and identifying operational gaps before customers are affected.

2. Customer Experience Score (Offline + Online)

- A survey-less measure of overall experience, inferred from observed behaviour, environmental readiness, and friction events.

3. Journey Friction Index (Offline + Online)

- A pinpointed view of which part of the journey — finding, deciding, paying, fulfilling, or supporting — caused dissatisfaction.

4. Dissatisfaction & Churn Prediction

- A forward-looking signal that identifies customers at risk of reducing visits, switching channels, or moving to competitors before they actually do.
- These outputs form the backbone of the predictive experience system and enable timely, targeted action.

SECTION 2 — OFFLINE STORES EXPERIENCE FUNDAMENTALS

Understanding experience in offline stores begins with the simplest, most observable signals and then expands into deeper behavioural and operational insights. The goal is to build upwards from store readiness, to in-store journey signals, to post-visit behaviour, feeding the core outputs: Store Readiness Score, Customer Experience Score, Journey Friction Index, and Dissatisfaction & Churn Prediction.

A. BASIC SIGNALS

These are foundational, operationally available signals that rely on existing POS data or simple process tools. They reflect first principles: if the store is not ready, every downstream customer signal becomes distorted.

Signal	How It Is Captured	KPIs / Metrics	Outcome (What It Tells Us)	Core Output(s)
RFQM (Recency, Frequency, Quantity, Monetary)	POS transaction history	Recency, visit frequency, average units, spend	Engagement level and early signs of churn	Customer Experience Score, Dissatisfaction & Churn Prediction
AOV and Basket Composition	POS basket data	AOV, mix, value shifts	Trade-down, premium loss, change in buying behaviour	Customer Experience Score, Churn Prediction
Store Readiness via Experience Manager App	Daily checklist app (AC, lighting, cleanliness, clutter, fixtures, staffing, maintenance issues)	Readiness %, issue recurrence	Establishes if the store was fit for customers that day	Store Readiness Score
Simple Conversion Proxy	POS bills per time-band	Conversion %, sales per visitor	Early indicator of friction or mismatch	Customer Experience Score

	(approximate walk-ins)			
Billing Time (Basic)	POS timestamps	Billing duration	Early detection of slow checkout	Journey Friction Index
Returns and Reason Codes	POS return records	Return %, reason clusters	Product dissatisfaction or mismatch	Customer Experience Score, Churn Prediction
Discount / Promotion Dependence	POS applied discounts	% of sales on discount	Value perception challenges	Customer Experience Score
Competitor Pricing (Key KVIs)	Price intelligence feeds or manual scans	Price index	Whether value perception is aligned before arrival	Store Readiness Score, Churn Prediction
Assortment Breadth and Depth	Store ranging files	SKU count by category	Assortment gaps affecting expectations	Store Readiness Score, Journey Friction Index

B. INTERMEDIATE SIGNALS

These require a structured process, instrumentation, or additional data collection, but remain operationally realistic.

Signal	How It Is Captured	KPIs / Metrics	Outcome (What It Tells Us)	Core Output(s)
Footfall and Walk-in vs Billing	Door counters	Walk-ins, conversion	Customers entering but not buying	Customer Experience Score, Journey Friction Index
Shelf Availability (Enhanced)	High-frequency staff scans	OOS minutes, SKU uptime	Persistent out-of-stock patterns	Store Readiness Score
Queue Length and Abandonment	Queue counters or basic overhead count	Queue time, abandonment	Checkout friction and staffing needs	Journey Friction Index, Customer Experience Score
Zone-Level Dwell (High-Level)	Overhead heatmaps (non-identity)	Dwell time by zone	Identifies confusing or ignored areas	Journey Friction Index
Staff Responsiveness	Staff app to log interactions	Assist time	Measures speed of customer assistance	Customer Experience Score
Staffing Allocation	Roster vs actual POS/staff activity	Counters open vs peak demand	Understaffing or misaligned staffing	Store Readiness Score, Journey Friction Index
Price Override / Promo Mismatch	POS override logs	Override count, mismatch frequency	Tells us where pricing or signage is unclear	Journey Friction Index

Store-Level Social Sentiment	Geo-tagged store mentions; social listening tools	Sentiment index, frequency of negative mentions	Independent validation of operational issues surfacing publicly	Customer Experience Score, Dissatisfaction & Churn Prediction

C. ADVANCED SIGNALS

These rely on deeper instrumentation, CV, ML, or identity stitching to reveal nuanced behavioural and operational patterns.

Signal	How It Is Captured	KPIs / Metrics	Outcome (What It Tells Us)	Core Output(s)
Detailed Path and Movement Loops	Full-store heatmap and path analytics	Loop frequency, time-to-find	Identifies navigation and search friction	Journey Friction Index
Pick–Putback and Hesitation Behaviour	Zone-level CV	Pick–putback %, hesitation	Pricing, quality, or information friction	Journey Friction Index, Customer Experience Score
Body-Language Friction Cues	Silhouette/posture CV (non-biometric)	Frustration index	Detects live dissatisfaction pockets	Customer Experience Score
Comfort Index (Temperature and Noise Mapping)	Distributed sensors	Comfort score	When discomfort affects dwell or conversion	Store Readiness Score
Cross-Store Switching	Identity stitching across stores	Store-switch rate	Customers are avoiding a specific store	Dissatisfaction & Churn Prediction
Offline to Online Migration	Omni-channel identity graph	Channel shift after store visits	Indicates unresolved in-store friction	Dissatisfaction & Churn Prediction

SECTION 3 — ONLINE EXPERIENCE FUNDAMENTALS

Understanding digital experience begins with basic signals already available in app, web, OMS, and CRM systems. These first-principles signals show whether customers can find what they want, complete their journey easily, and receive reliable fulfilment. Intermediate and advanced signals build on these foundations to reveal deeper friction patterns, sentiment, and predictive risk.

A. BASIC SIGNALS

Signal	How It Is Captured	KPIs / Metrics	Outcome (What It Tells Us)	Core Output(s)
RFQM	Digital transaction history	Recency, frequency, units, spend	Customer engagement and early churn indicators	Customer Experience Score, Dissatisfaction & Churn Prediction
AOV and Basket Composition	Cart and order logs	AOV, category mix	Shifts in value perception or buying behaviour	Customer Experience Score
Funnel Drop-off (View → Cart → Checkout → Payment)	Clickstream analytics	Drop-off %, conversion progression	Where friction occurs in the digital journey	Journey Friction Index
Payment Failures and Retries	Payment gateway logs	Failure %, retry %, fallback attempts	Trust or payment UX friction	Journey Friction Index, Customer Experience Score
Delivery Performance	OMS + last-mile feeds	On-time %, delays, cancellations, substitutions	Reliability and quality of fulfilment	Customer Experience Score, Dissatisfaction & Churn Prediction
Returns and Reason Codes	Digital returns workflow	Return %, reason clusters	Product/content mismatch	Customer Experience Score
Search Performance	Search logs	Zero-result %, search-to-conversion	Catalogue and tagging quality	Journey Friction Index
Out of Stock (OOS)	Real-time catalogue availability	OOS %, SKU uptime	Immediate friction due to unavailability	Journey Friction Index, Customer Experience Score
Assortment Range	Category listing completeness	Category coverage %, missing key SKUs	Whether the digital store meets customer expectations	Store Readiness Score, Journey Friction Index
Competitor Pricing	Price comparison feeds on key SKUs	Price index, price gaps	How value perception is shaped before checkout	Customer Experience Score

Promo / Coupon Issues	Checkout logs	Promo success %, failure %, invalid attempts	Pricing or promotion clarity issues	Journey Friction Index
App and Web Performance	Telemetry & web vitals	Load time, crash rate, and responsiveness	Technical friction affecting engagement	Customer Experience Score
Add-to-Cart vs Purchase	Cart activity vs orders	Cart abandonment %, hesitation	Pricing, delivery, UX or trust issues	Customer Experience Score
Customer Service Touchpoints (Chat/Bot/WhatsApp)	CRM and chat logs	Contact rate, escalation rate	Themes driving friction or confusion	Journey Friction Index
Support Tickets	CRM systems	Ticket volume, categories, and resolution time	Where digital journeys fail operationally	Customer Experience Score
Ratings and Reviews	App/web review systems	Avg rating, rating distribution	Customer sentiment on products and service	Customer Experience Score
Store-Pick Order Quality	OMS pick logs	Pick accuracy, pick time	Offline readiness affecting online fulfilment	Customer Experience Score
Social Media Sentiment	Public posts referencing digital experience	Sentiment index, spikes	External validation of digital experience quality	Customer Experience Score

B. INTERMEDIATE / ADVANCED SIGNALS

Signal	How It Is Captured	KPIs / Metrics	Outcome (What It Tells Us)	Core Output(s)
Behavioural Micro-Journey Patterns	Session path analytics	Micro-conversions , friction clusters	Predicts dissatisfaction before drop-off occurs	Customer Experience Score, Journey Friction Index
NLP-Based Sentiment from Chats, Emails, Reviews	NLP/ML text analysis	Sentiment index, dominant themes	Hidden friction not seen in clickstream data	Customer Experience Score
Product Content Quality Issues	ML on product descriptions and images + return reasons	Content mismatch %, conversion correlation	When inaccurate/incomplete content impacts purchase confidence	Journey Friction Index

Personalisation Failure Signals	CTR patterns, relevance scoring	Recommendation CTR %, relevance score	When recommendations degrade experience	Customer Experience Score
Delivery Experience Prediction	ML using SLA history, traffic, and weather	Predicted SLA breach likelihood	Proactive intervention before a poor experience occurs	Customer Experience Score
Channel Switching (Online ↔ Offline)	Identity stitching across systems	Channel shift %, friction escape behaviour	When dissatisfaction in one channel pushes customers to another	Dissatisfaction & Churn Prediction

SECTION 4 — CORE OUTPUTS

All offline and online signals ultimately consolidate into four core outputs. These outputs provide a simple, consistent way to understand daily readiness, infer customer experience without asking, identify where friction occurred, and predict which customers are at risk. Each score has a clear calculation method and a clear purpose.

Output	How the Score Is Calculated (Inputs)	What the Score Means (Interpretation)
Store Readiness Score	AC/temperature, lighting, cleanliness, staffing presence, assortment completeness, out-of-stock checks, pricing consistency, Experience Manager App logs, maintenance issues	Indicates whether the store was operationally ready to serve customers. Forms the baseline for interpreting all in-store behaviour.
Customer Experience Score	RFQM patterns, basket shifts, queue and billing performance, app/web responsiveness, payment reliability, search effectiveness, fulfilment quality, returns, customer service interactions, ratings and sentiment	A survey-less measure of how customers actually experienced their journey across stores and digital channels.
Journey Friction Index	Search zero-results, funnel drop-offs, cart abandonment factors, promo/coupon failures, dwell/navigation delays, queue abandonment, pricing conflicts, fulfilment issues, support escalation patterns	Shows exactly where friction occurred — whether in finding, deciding, paying, receiving, or seeking support.
Dissatisfaction & Churn Prediction	Declining RFQM, reduced visit frequency, shrinking basket size, repeated returns, channel switching after friction, fulfilment delays, ongoing negative sentiment	Predicts which customers are likely to disengage or reduce spend, enabling proactive action before churn occurs.

SECTION 5 — ACTIONS & FEEDBACK LOOPS

Signals and scores are only useful when they drive timely and consistent action. This section explains how each core output triggers specific responses across store operations, digital product, fulfilment, pricing, and customer service. It also describes how every action feeds back into the system to improve accuracy and performance over time.

ACTIONS & FEEDBACK MATRIX

Core Output	What Triggers Action	Action Taken	Owner	Feedback Loop (What Is Re-measured)
Store Readiness Score	Low readiness score (e.g., AC, lighting, cleanliness, staffing gaps, assortment gaps, pricing inconsistencies)	Fix issues immediately (environment, maintenance, stock, staffing). Mark resolution in Experience Manager App.	Store Manager, Experience Team	Readiness score re-evaluated; store condition validated through next scheduled check or visit.
Customer Experience Score	Low or declining score driven by behaviour (e.g., payment failures, delivery delays, long queues, poor app performance, rising returns)	Identify top drivers by channel (store, digital, fulfilment) and initiate corrective actions such as process fixes, staffing changes, content corrections, UX improvements.	CX Team, Store Ops, Product, Fulfilment	Recalculate CX score after fixes; track whether engagement and RFQM recover.
Journey Friction Index	High friction detected at a specific journey step (search, navigation, checkout, payment, fulfilment, support)	Deploy targeted fixes: improve search tagging, correct pricing or promos, reduce queue time, resolve checkout issues, fix delivery routes, clarify content.	Product, Merchandising, Store Ops, Customer Service	Measure reduction in friction signals at the same journey step.
Dissatisfaction & Churn Prediction	Customer or household flagged as high-risk based on behaviour trends (declining frequency, smaller baskets, complaints, negative sentiment, fulfilment issues)	Proactive retention and recovery actions — personalised outreach, issue resolution, cross-channel support, service guarantees, or targeted offers.	CRM, CX Team	Track whether RFQM improves and churn risk decreases after intervention.

How the System Learns

The cycle of **signals → outputs → actions → feedback** ensures that the entire experience engine improves over time.
Once actions are taken:

- The same signals are measured again
- Scores update automatically
- Improvements (or lack of improvement) are visible
- Teams can adjust or escalate
- Models refine their weighting of signals based on observed outcomes

This creates a **self-correcting feedback loop** where both operations and predictions get better with each cycle.